EXP. NO: 14
DATE : 15/10/26

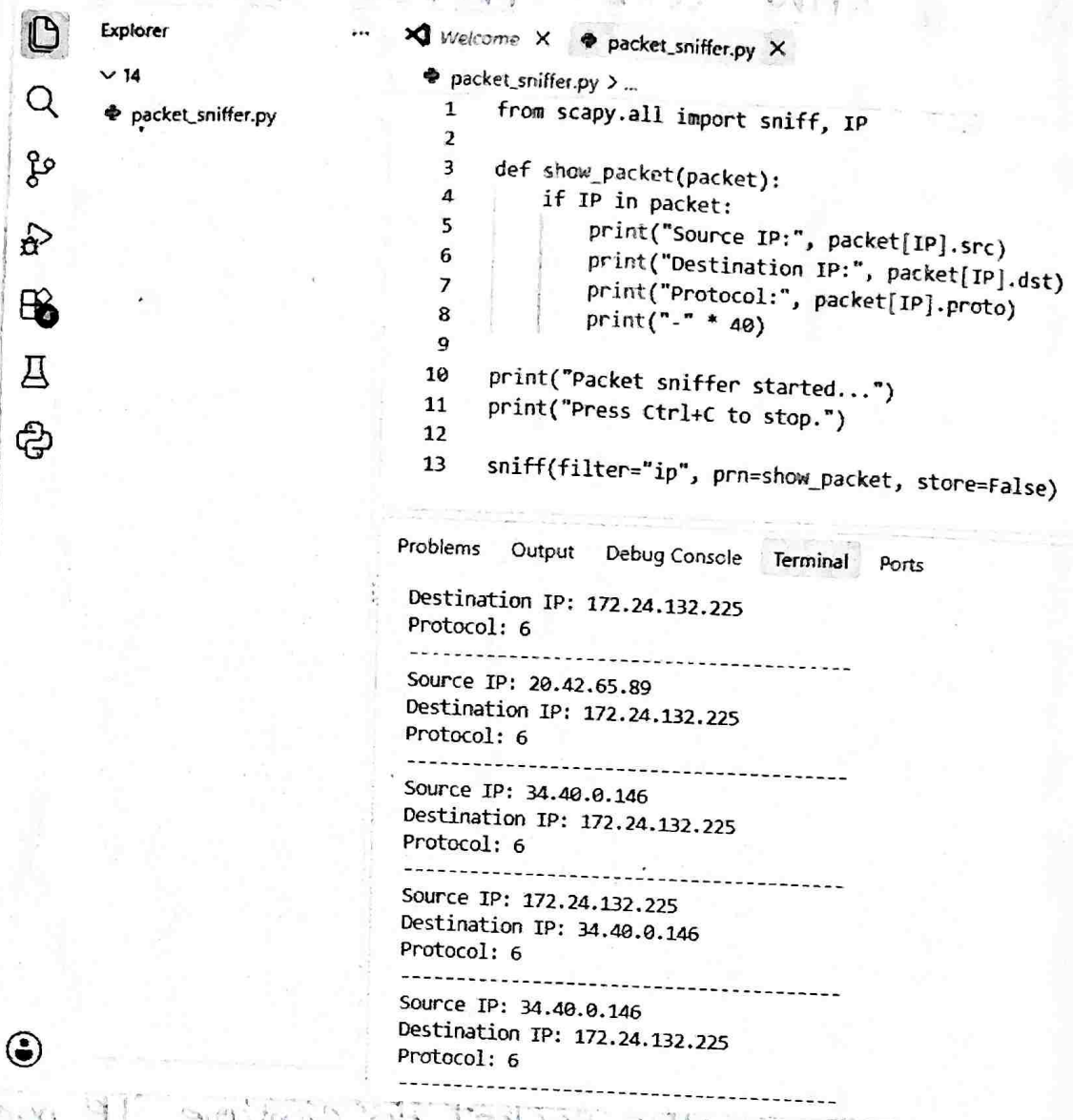
RAW SOCKET PACKET SNIFFING

AIM:

To Implement a packet sniffing program using RAW Sockets and capture basic Information from network packets.

PROCEDURE :

1) Open VS Code & create a Python file named packet_sniffer.py
2) Import the required Socket module.
3) Create a RAW socket for receiving network packets.
4) Configure the socket to capture IP packets.
5) Start the packets sniffing process.
6) Continuously listen for incoming packets.
7) When a packet is received, extract information from its IP header.
8) Extract the following information :
   Source IP address, Destination IP address
   IP version, TTL, Protocol
9) Display the extracted packet information on the screen.


Explorer
14
packet_sniffer.py
Welcome
packet_sniffer.py
packet_sniffer.py > ...
1 from scapy.all import sniff, IP
2
3 def show_packet(packet):
4 if IP in packet:
5 print("Source IP:", packet[IP].src)
6 print("Destination IP:", packet[IP].dst)
7 print("Protocol:", packet[IP].proto)
8 print("-" * 40)
9
10 print("Packet sniffer started...")
11 print("Press Ctrl+C to stop.")
12
13 sniff(filter="ip", prn=show_packet, store=False)
Problems
Output
Debug Console
Terminal
Ports
Destination IP: 172.24.132.225
Protocol: 6
Source IP: 20.42.65.89
Destination IP: 172.24.132.225
Protocol: 6
Source IP: 34.40.0.146
Destination IP: 172.24.132.225
Protocol: 6
Source IP: 172.24.132.225
Destination IP: 34.40.0.146
Protocol: 6
Source IP: 34.40.0.146
Destination IP: 172.24.132.225
Protocol: 6

10) Generate network traffic from another terminal using

ping 8.8.8.8.

11) Observe the packets captured by the sniffer.

12) Continue capturing packets for the required duration.

13) Stop the packet sniffing program using Ctrl + C.

14) Analyze the captured packet information.

RESULT:

Thus a RAW socket based packet sniffing program was successfully implemented.